THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solid sphere of radius 9 cm is completely dropped into water in a cylindrical vessel of base radius 6 cm, and no water spills. The rise in the water level is: 1

(a) 54 cm **(b)** 13.5 cm **(c)** 27 cm **(d)** 1.5 cm

2. A capsule is a cylinder of radius 21 cm and length 14 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 4620 cm² **(b)** 14784 cm² **(c)** 10164 cm² **(d)** 7392 cm²

3. A toy is a cone of base radius 35 cm and height 84 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 16940 cm² **(b)** 10010 cm² **(c)** 21560 cm² **(d)** 17710 cm²

4. A solid metal cuboid 24 cm by 24 cm by 12 cm is melted and recast into cubes of edge 3 cm. The number of cubes obtained is: 1

(a) 768 **(b)** 256 **(c)** 512 **(d)** 255

5. A tent is a cylinder of radius 7 m and height 12 m surmounted by a cone of the same radius and height 24 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 1056 m² **(b)** 550 m² **(c)** 528 m² **(d)** 1078 m²

6. A hemispherical bowl has outer radius 12 cm and inner radius 3 cm. The volume of material used to make the bowl is: 1

(a) 1782 cm³ **(b)** 7128 cm³ **(c)** 10692 cm³ **(d)** 3564 cm³

7. A hemispherical depression of diameter 84 cm, the largest possible, is scooped out of the top face of a cubical block of edge 84 cm. The volume of the remaining block is: 1

(a) 437472 cm³ **(b)** 592704 cm³ **(c)** 282240 cm³ **(d)** 155232 cm³

8. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 4 cm. The length of the wire is: 1

(a) 576 cm **(b)** 144 cm **(c)** 72 cm **(d)** 288 cm

9. A solid sphere of radius 12 cm is completely dropped into water in a cylindrical vessel of base radius 12 cm, and no water spills. The rise in the water level is: 1

(a) 16 cm **(b)** 32 cm **(c)** 8 cm **(d)** 1 cm

10. A capsule is a cylinder of radius 3.5 cm and length 20 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 1188 cm² **(b)** 594 cm² **(c)** 671 cm² **(d)** 517 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A solid is a cube of edge 56 cm surmounted by a hemisphere of diameter 56 cm (the hemisphere rests on a face of the cube). Find the total surface area of the solid. 2

12. A solid metallic cone of base radius 18 cm and height 9 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

13. A solid metallic cylinder of base radius 12 cm and height 24 cm is melted and recast into solid spheres of radius 3 cm. Find the number of spheres obtained. 2

14. A solid is a cube of edge 63 cm surmounted by a hemisphere of diameter 63 cm (the hemisphere rests on a face of the cube). Find the total surface area of the solid. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A capsule is a cylinder of radius 10.5 cm and length 25 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

16. A toy is a cone of base radius 21 cm and height 21 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

17. A wooden article is made by scooping out a hemisphere of radius 10.5 cm from each end of a solid cylinder of height 20 cm and base radius 10.5 cm. Find the total surface area of the article. 3

OR

A capsule is a cylinder of radius 21 cm and length 8 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

18. A toy is a cone of base radius 10.5 cm and height 42 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

OR

A wooden article is made by scooping out a hemisphere of radius 14 cm from each end of a solid cylinder of height 25 cm and base radius 14 cm. Find the total surface area of the article. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A conical vessel of radius 9 cm and height 8 cm is full of water. The water is poured into a cylindrical vessel of base radius 3 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 7 cm and height 30 cm is poured into an empty rectangular tank of base 30 cm by 22 cm. Find the height of water in the rectangular tank. 5

20. A conical vessel of radius 9 cm and height 16 cm is full of water. The water is poured into a cylindrical vessel of base radius 6 cm. Find the height of the water in the cylindrical vessel. 5

OR

Water from a full cylindrical tank of radius 7 cm and height 15 cm is poured into an empty rectangular tank of base 44 cm by 21 cm. Find the height of water in the rectangular tank. 5

END OF QUESTION PAPER — Total: 40 Marks